

HLA regions from 610 assembled haplotypes: extraction, typing and pangenome views

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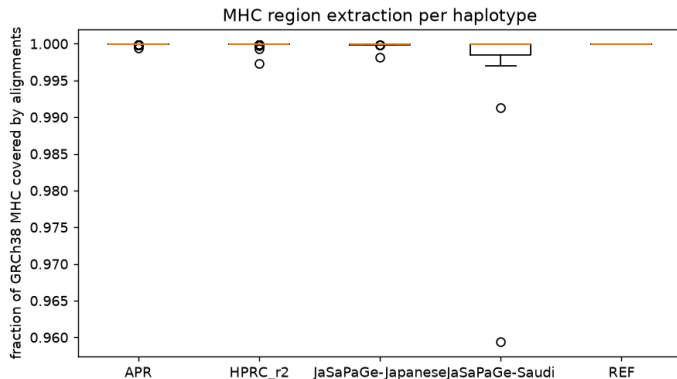
BioHackathon Japan 2026, HLA/pangenome group

15 September 2026

One CWL workflow: pgr-tk extraction, Immuannot typing, pgr-tk and pggg graphs

- ▶ 610 haplotypes: APR (UAE Arab) 106, HPRC release 2 464, JaSaPaGe Saudi 18 and Japanese 20, GRCh38, CHM13
- ▶ MHC (GRCh38 chr6:28.5–33.5 Mb, 100 kb flank) pulled from each assembly with pgr-tk pgr-query, about 30 s per genome
- ▶ Immuannot (IPD-IMGT/HLA 3.55) gives gene structure and full-resolution allele calls per haplotype
- ▶ Per gene (19 genes): sequences cut at the Immuannot coordinates, pgr-tk principal-bundle plots, pggg graphs rendered with odgi
- ▶ Whole MHC and the class II block (DRA to DMA): pgr-tk bundle decomposition, dendrograms, bundle graphs
- ▶ Runs on the NIG node without containers (Toil, --no-container); repo github.com/leechuck/pangenome-bh26

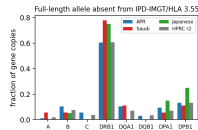
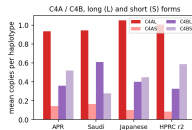
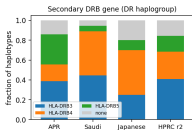
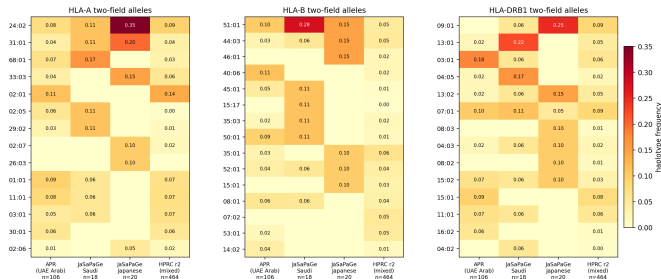
The MHC comes out intact from 608 of 610 haplotypes



- y: fraction of the GRCh38 MHC covered by the extracted segments; median 1.000 in every cohort, 599 haplotypes on a single contig
- Five 1000G JPT individuals were assembled by both HPRC and JaSaPaGe: 59 of 61 full-resolution gene calls agree, the two exceptions differ only in an intron

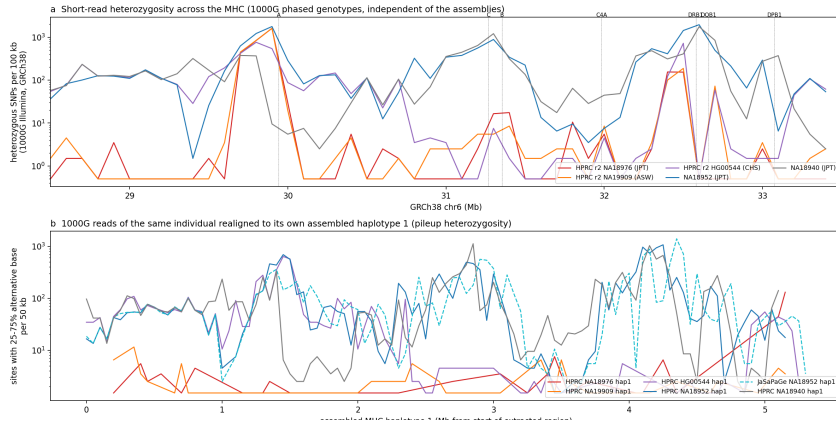
Allele frequencies separate the cohorts; DRB1 full-length sequences are mostly new

HLA allele landscape across 608 assembled haplotypes (Immuannot on assembly-derived MHC)



Immuannot consensus calls, two-field alleles; novel = full-length sequence absent from IPD-IMGT/HLA 3.55

Two HPRC individuals are truly MHC-homozygous; one JaSaPaGe assembly carries the same haplotype twice

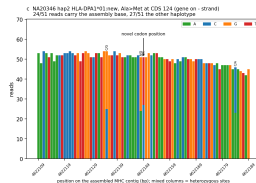
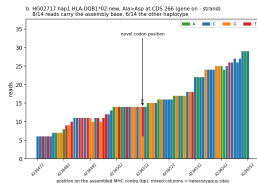


- NA18976, NA19909 (red, orange): under 100 heterozygous 1000G SNPs across 5 Mb, matching their two near-identical assembled haplotypes; JaSaPaGe NA18952 (dashed): own reads show full heterozygosity on both “haplotypes”, hap1 = hap2

Substitution-type novel coding alleles hold up in the individual's own reads; indel-only ones do not

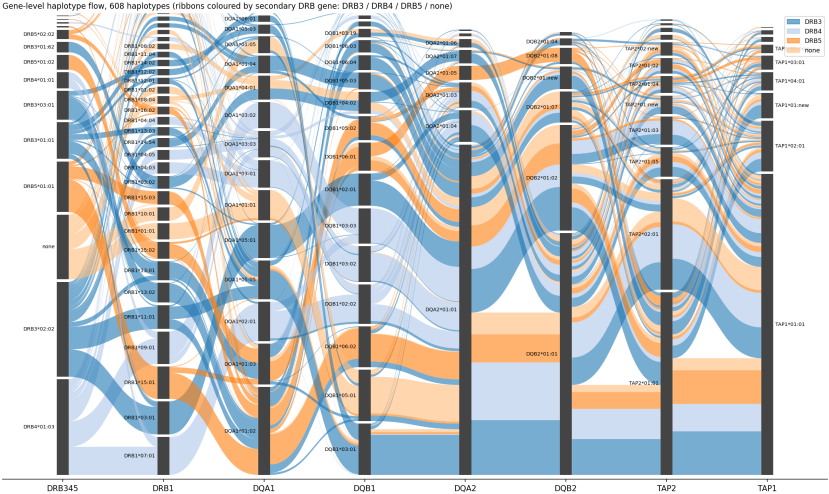
a Candidate novel coding alleles (ImmunoNot 'new' with CDS differences) in classical HLA genes. private = variant 31-mer absent from all other 609 haplotypes; red = differences include indels (typical HiFi homopolymer assembly errors, treat as artefacts); green = heterozygous support in the individual's own short reads

haplotype	cohort	gene	closest known allele	coding difference(s) (known > assembly)	private variants	also on other hap	own 1000G Illumina reads (realigned to assembly)
apr003#1	APR	A	A*24:02:102:01	4 subst.: Val>Leu, Phe>Tyr, Ser>Pro, Lys>Asn	1	no	no public reads
apr011#1	APR	C	C*03:02:02:01	3 subst.: Ile>Leu, Ala>Gly, Gly>Cys	0	no	no public reads
apr001#1	APR	DRB1	DRB1*16:01:01:01	5 subst.: Gln>His, Val>Val, Leu>Arg, Met>Val, Arg>Arg	0	no	no public reads
apr038#1	APR	DRB1	DRB1*14:06:01	5 subst. + indels: Arg>Lys, Val>Ala, Arg>Arg, Cys>Cys, His>Gln	1	no	no public reads
apr002#1	APR	DQB1	DQB1*02:02:01:01	1 subst.: Leu>Leu	0	yes	no public reads
apr015#2	APR	DPB1	DPB1*10:01:01:01	1 subst.: Leu>Leu	1	no	no public reads
HG02717#1	HPRC r2	DQB1	DQB1*02:02:01:01	1 subst.: Ala>Asp	0	no	8/14 assembly base, 6 other hap
NA20346#2	HPRC r2	DPA1	DPA1*01:03:01:18Q	1 subst.: Ala>Met	1	no	24/51 assembly base, 27 other hap
ksa004#1	JaSaPaGe-Saudi	A	A*02:01:01:134Q	3 indels only (1-bp deletion, 2 insertions)	-	-	no public reads
HG03516#2	HPRC r2	DPA1	DPA1*03:01:01:01	1-bp insertion only	-	-	-



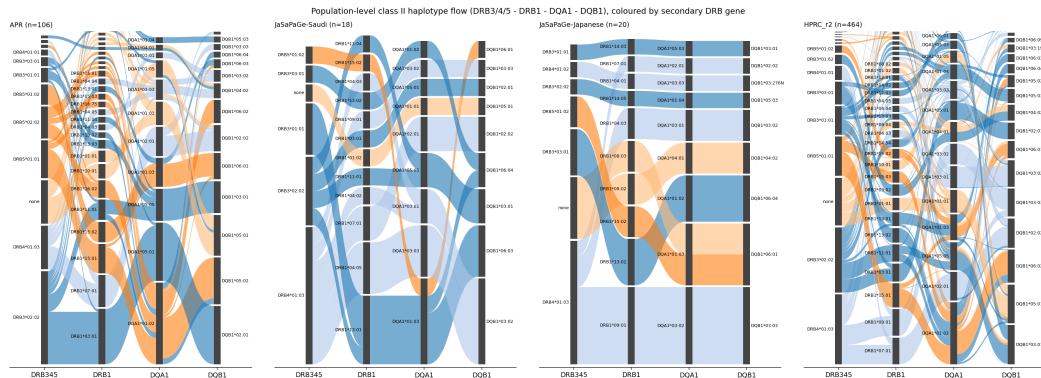
Pileups: the individual's 1000G Illumina reads realigned to its assembled contig; mixed columns = heterozygous sites. APR and JaSaPaGe reads are not public

425 distinct gene-level class II haplotypes among 608; secondary DRB gene structures the flow



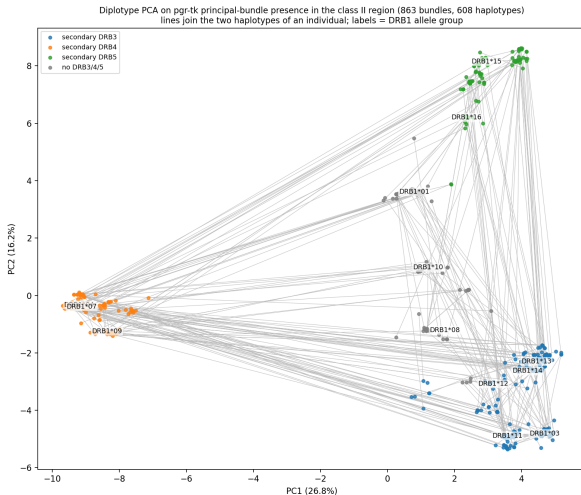
Two-field alleles per haplotype; ribbons coloured by DRB3 / DRB4 / DRB5 / none. After Chin, ASHI 2023 (94 haplotypes)

The DR-DQ block differs by population



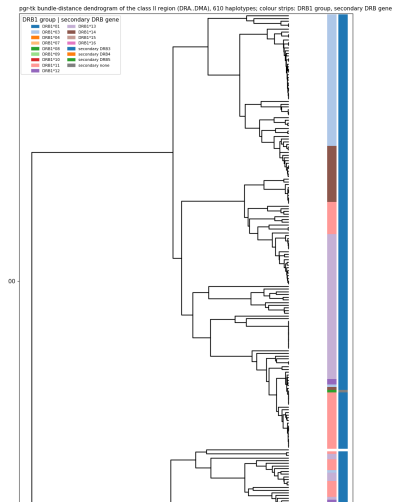
- Saudi: DRB4 haplotypes (DR7, DR4) 44%; Japanese: DRB1*09:01–DQB1*03:03 and DRB1*15:02–DQB1*06:01; APR: DRB1*03:01–DQB1*02:01 18%

Sequence-level structure of the class II region follows the DR haplogroups



PCA on presence of 863 pgr-tk principal bundles (DRA to DMA); each point one haplotype, grey lines join the two haplotypes of an individual

Bundle-distance clusters are near-pure DRB1 allele groups

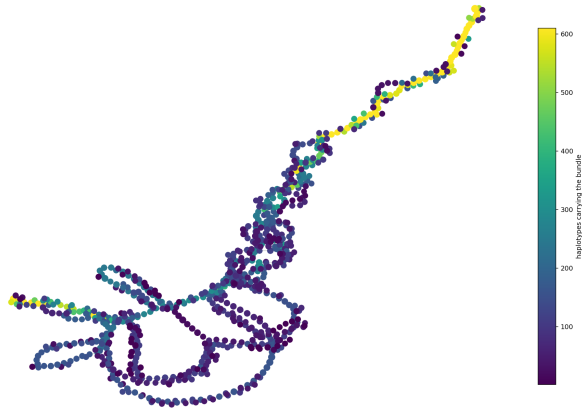


Top of the pgr-tk bundle-distance dendrogram of 610 class II haplotypes; strips: DRB1 group, secondary DRB gene. Full tree:

fig9_classII_dendrogram.png

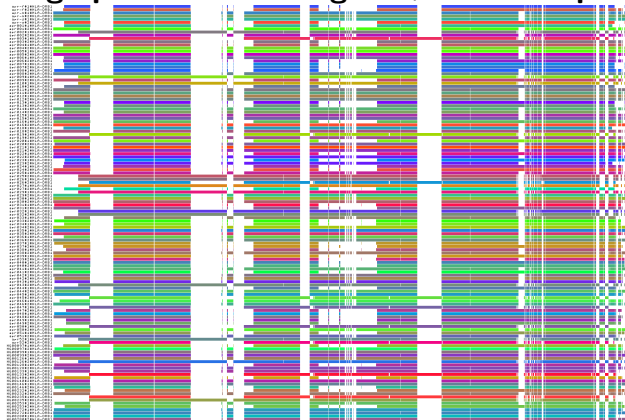
The class II bundle graph is linear except for the DRB block

pgprtk principal-bundle graph of the class II region (DRA, DMA), 630 haplotypes, 863 bundles, 1341 adjacencies
node size = bundle length, colour = number of haplotypes; edge width = haplotypes sharing the adjacency



863 principal bundles, node size = bundle length, colour = haplotypes carrying it, edges = consecutive bundles on a haplotype

Per-gene pangenome graphs for 19 HLA genes, all 610 haplotypes



- ▶ HLA-DRB1 pggb graph (odgi viz, top rows of 610): rows are haplotypes, x is the graph, colour = path coverage
- ▶ Also per gene: pgr-tk bundle plot (SVG/HTML), dendrogram, MAP-graph GFA; whole-MHC bundle plot

Take-home

- ▶ Assembly-based HLA typing across three pangenome projects reproduces known population frequencies and reveals data problems (duplicated haplotype, homozygous individuals) that short reads alone would not
- ▶ 60–78% of DRB1 gene copies have no full-length match in IPD-IMGT/HLA; Arab haplotypes (APR + Saudi, 124) are a submission-ready resource once read support is added
- ▶ pgr-tk bundle decomposition recovers DR haplogroup structure without any allele calls; scales from 94 to 610 haplotypes
- ▶ Open: APR and JaSaPaGe raw reads for novel-allele validation; KIR and IGH with the same workflow; HPRC per-population split
- ▶ Workflow, scripts, figures and tables: github.com/leechuck/pangenome-bh26 (h1a/)